HAWK Fakultät Ingenieurwissenschaften und Gesundheit Göttingen

POLITECNICO MILANO 1863

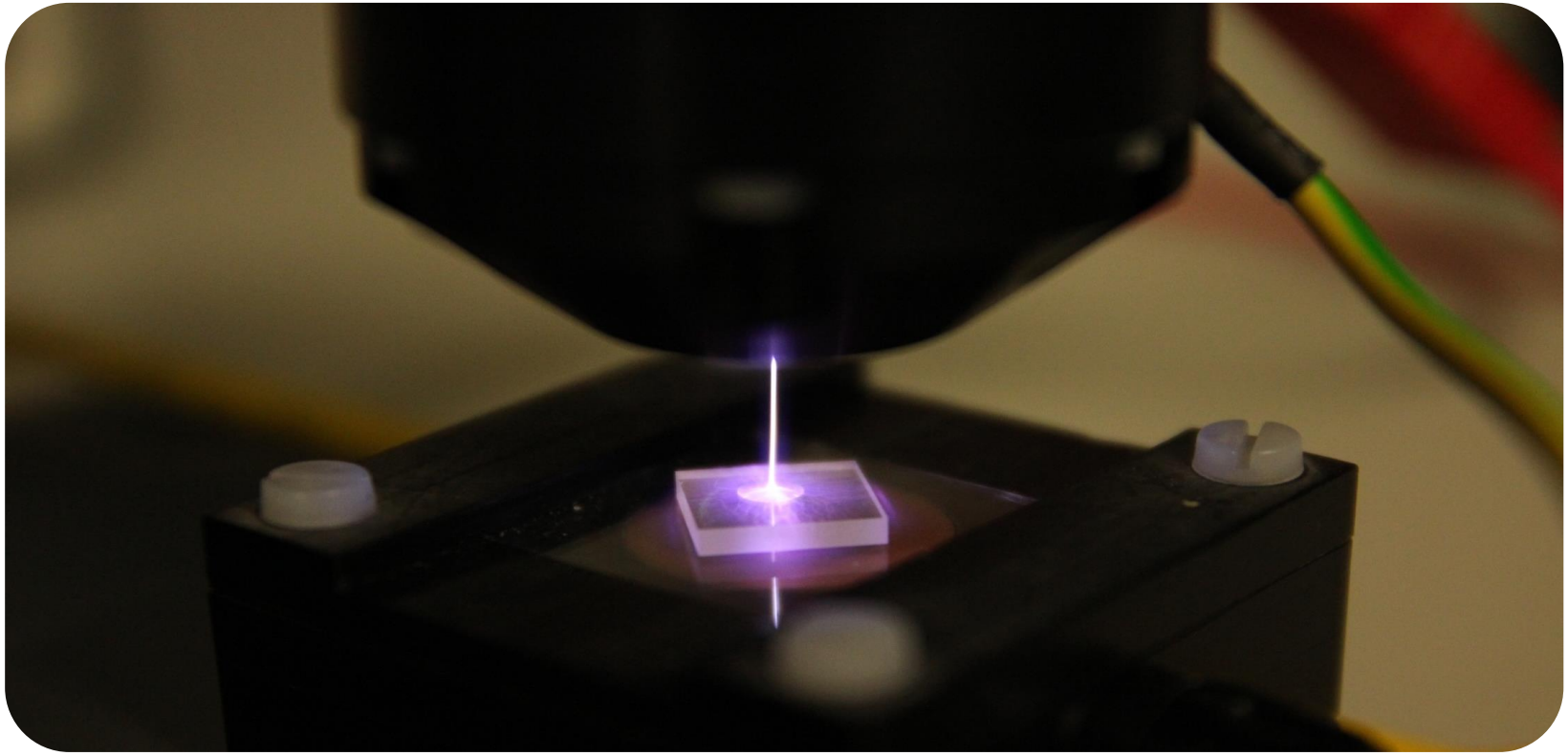
Repetition to

ADVANCED TOPICS IN ENGINEERING PHYSICS

‘Plasma and its applications’

Prof. Dr. Christoph Gerhard, 2024

2 Plasma discharges and sources



- 46. Name at least three effects that may cause deviations from thermodynamic equilibrium within plasmas.
- 47. In which case does the electron energy distribution function not correspond to the Maxwell distribution?
- 48. What is the basic assumption of the so-called Druyvesteyn distributio?
- 49. What is the reason for the extremely non-uniform population of energy levels in non-equilibrium plasmas?
- 50. What is an optically ‘thick plasma’?
- 51. Name an example of a plasma, where the mean electron temperature is higher than the translation temperature of heavy particles.
- 52. What is the ‘translation temperature’ of a plasma?

- 53. Why do plasmas feature higher vibrational temperature than rotational temperature?
- 54. Name some loss processes that may occur within a plasma.
- 55. Name the three most important types of DC plasma discharges
- 56. Give a selection of – about three – parameters that can be used for the classification of plasmas.
- 57. What is the main characteristics of isothermal plasmas?
- 58. What is a Townsend discharge?
- 59. Name two mechanisms, which lead to the generation of the initial electron of a Townsend discharge.
- 60. Which characteristic parameter describes the number of ionisation events per electron and distance?

- 61. How to determine the energy of an electron between two collisions?
- 62. What are the main influencing parameters for the determination of the number of electrons generated via avalanche ionisation?
- 63. What is the so-called ‘reduced field strength’?
- 64. Why do we usually not consider the Townsend coefficient β?
- 65. Give a short definition of the second Townsend coefficient.
- 66. Which conditions need to be fulfilled for a self-maintaining ignition of a discharge within a gas-filled gap?
- 67. Give a short definition of Paschen’s law.
- 68. Draw a qualitative sketch of a Paschen’s curve…do not forget to label both the y- and the x-axis.

- 69. What is the so-called ‘cathode fall’?
- 70. What is the consequence of field emission of electrons caused by the energy transfer of incident ions?
- 71. What is a ‘glow discharge’?
- 72. A glow discharge comes along with the formation of characteristic dark and bright zones between the electrodes. Name at least three of these zones.
- 73. Which type of discharge features a higher discharge current, the glow discharge or the anomalous glow discharge?
- 74. What is the main difference of a Townsend discharge and a glow discharge?
- 75. Define ‘non-self-maintaining’ and ‘self-maintaining’.

- 76. What does the Child-Langmuir-law indicate?
- 77. When does thermionic emission of charge carriers occur?
- 78. Which type of plasma is driven by thermionic emission?
- 79. Further increase in voltage of an anomalous glow discharge may lead to the formation of an arc discharge. What happens to the current density and the voltage in this case?
- 80. Name at least two examples for arc discharges.
- 81. What is – approximately – the current density within an isothermal arc discharge?
- 82. Which mechanism supports the (spatial) constriction of an arc plasma occur?
- 83. Give the typical temperature range of arc discharges.

- 84. What is the minimum temperature for thermionic emission of electrons?
- 85. Which equation can be applied for the determination of current density during thermionic emission?
- 86. What is the Schottky effect?
- 87. What is a plasma filament?